

Anoop Gupta

AI/ML Engineer & Full-Stack Developer

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<https://github.com/AnoopG7> | <https://anoopg-portfolio.vercel.app/>

PROFESSIONAL SUMMARY

AI/ML Engineer and Full-Stack Developer with hands-on experience building production-grade AI systems, RAG architectures, multi-agent workflows, and scalable SaaS platforms using FastAPI, React, TypeScript, PostgreSQL, and modern LLM tooling. Skilled in LLM applications, backend architecture, cloud deployment, and production reliability. Passionate about building intelligent products with strong backend architecture, performance optimization, and production reliability.

Skills

AI / Machine Learning - Python, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Hugging Face, LangChain, RAG, Multi-Agent Workflows, Prompt Engineering, Model Evaluation, Semantic Search, Vector Embeddings

Backend Development - FastAPI, Node.js, Express.js, REST APIs, WebSockets, Socket.io, JWT Authentication, Async Processing, API Integration

Frontend Development - React, Next.js, TypeScript, JavaScript, Tailwind CSS, Zustand, React Query, Shadcn UI, MUI, React Hook Form

Databases - PostgreSQL 16, MongoDB, MySQL, Supabase, Firebase, Database Schema Design

DevOps - Docker, Docker Compose, Kubernetes, Git, GitHub, GitHub Actions, CI/CD, Linux, Jenkins, Nginx, Swagger/OpenAPI, Prometheus, Grafana, ELK Stack

System Design - Distributed Systems, Scalable Backend Architecture, Microservices, Event-Driven Architecture, Message Queues, Caching, Load Balancing, API Design

Cloud - Terraform, AWS (EC2, S3, IAM, VPC, Auto Scaling, Elastic Load Balancer, CloudWatch)

Experience

AI/ML Engineer Intern | Shovihini Tech LLP

- Built AI-powered applications using LLMs, RAG pipelines, vector embeddings, and NLP workflows
- Developed scalable backend services for AI inference and data processing using Python and FastAPI
- Improved retrieval accuracy, latency, and response quality through prompt and pipeline optimization

Web Development Intern | CodeDev Verse

- Developed responsive full-stack web applications using modern frontend and backend technologies
 - Built APIs, integrated databases, and implemented authentication workflows
 - Contributed to debugging, testing, deployment, and performance optimization
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FEATURED PROJECTS

TSRL - Trading Strategy Research Lab

Tech Stack: React, TypeScript, FastAPI, Python, Scikit-learn, Yahoo Finance, Alpha Vantage

- Engineered 116 ML features for signal generation using Random Forest & Gradient Boosting, achieving 50+ risk metrics (Sharpe, Sortino, Max Drawdown, VaR, CVaR) per run.
- Built walk-forward validation that exposed 3 overfitted strategies as curve-fitting noise. Genetic Algorithm optimization converging 10x faster than Grid Search on large parameter spaces.
- Dual backtesting engines (event-driven + vectorized) with multi-source data pipeline and AI sentiment analysis. FastAPI backend, 20+ endpoints, interactive React dashboard.

AI Lead Gen Platform | AI-Powered Lead Generation SaaS

Tech Stack: FastAPI, PostgreSQL, React, Docker, OpenAI, Meta Graph API

- Built a deterministic AI agent using raw OpenAI function calling with 8 tools and a 10-turn autonomous execution loop
- Developed async FastAPI backend with custom JWT auth, CRM dashboard, and PostgreSQL schema
- Reduced inference costs by ~60% using Groq Llama-3 with 429 retry backoff, and JSON recovery

TutorX | AI-Powered RAG Tutoring System

Tech Stack: FastAPI, React, Supabase, pgvector, Groq

- Built a RAG tutor using NCERT textbooks for curriculum-aligned, context-aware explanations
- Designed 3-stage retrieval (pgvector search, query reformulation, chunk enrichment), reducing API calls from 14+ to 1
- Implemented PDF ingestion, embeddings generation, authentication, and Telegram bot integration

TaskFlow | AI-Powered Proactive Task Management

Tech Stack: FastAPI, React, Supabase, Telegram Bot

- Built a multi-action AI agent capable of executing 10 task operations atomically in a single LLM call
- Developed self-improving time estimation with bias tracking across the last 30 completed tasks
- Built APScheduler workflows for daily briefings, hourly alerts, and multi-channel task management

Portfolio → anoopg-portfolio.vercel.app

EDUCATION - Pursuing Bachelor of Technology (B.Tech) in Computer Science Engineering - AI/ML Specialization

ITM Skills University, Navi Mumbai, India | Graduation: 2028 | CGPA: 9.55/10

Achievements and Certifications

- Academic Excellence Award
- AWS Cloud Practitioner Essentials
- Machine Learning Specialization